

Table 1: Key open source tools in modern data infrastructure

Layer	Purpose	Popular open source tools
Data ingestion	Collect and stream data from sources	Apache Kafka, Apache NiFi, Fluentd, Logstash, Debezium
Data storage	Store structured/unstructured data	Hadoop HDFS, Apache Iceberg, Delta Lake, Apache Hudi
Data processing	Batch and real-time computation	Apache Spark, Apache Flink, Apache Beam
Orchestration and workflow	Schedule and manage data pipelines	Apache Airflow, Dagster, Prefect
Data transformation	Model and transform raw data	dbt (Data Build Tool), Apache Calcite
Query engines	Run SQL-like queries on Big Data	Presto, Trino, Apache Druid, ClickHouse
Monitoring and observability	Monitor system health and performance	Prometheus, Grafana, OpenTelemetry
Metadata and governance	Catalogue, lineage, and compliance tracking	Apache Atlas, Amundsen, OpenLineage
Visualisation and BI	Visualise data for analysis and decision-making	Apache Superset, Metabase, Redash
Machine learning/AI	Model training, tracking, and deployment	MLflow, Kubeflow, Hugging Face Transformers

array of open source tools, each specializing in a particular layer of the infrastructure. Table 1 outlines these tools and their primary purposes.

Real-world adoption and success stories

Many tech giants and data-driven companies have built their data infrastructure on open source tools.

- **Netflix** uses Apache Kafka, Spark, and Iceberg to handle petabytes of data for streaming recommendations and analytics.
- **Uber** built its Michelangelo ML platform with components like Kafka, Hadoop, and Flink.
- **Airbnb** uses Presto, Airflow, and Superset for analytics and data science workflows.
- **LinkedIn**, the birthplace of Kafka, uses an entire open source-driven stack to power real-time data pipelines and personalised content delivery.

These companies not only consume open source but also contribute back, enhancing the ecosystem for everyone.

Challenges of open source in data infrastructure

Despite the benefits, open source is not without challenges.

Talent requirements: Managing and customising open source tools requires skilled engineers.

Operational complexity: Running distributed systems like Spark, Kafka, or Flink at scale demands expertise in DevOps, monitoring, and security.

Lack of official support: Unless paired with a vendor (e.g., Confluent for Kafka, Databricks for Spark), support

may be community-driven, which can delay resolution of critical issues.

Fragmentation: With so many tools available, assembling a cohesive stack requires careful planning and integration.

These challenges, however, are being addressed by a growing ecosystem of commercial support and managed cloud services (e.g., AWS EMR, GCP Dataproc, Azure Data Factory) built around open source.

The future of open source in data infrastructure

The future is increasingly open, modular, and interoperable. Some trends to watch:

Open standards: Projects like Iceberg and Delta Lake are pushing open table formats for lakehouses.

Data mesh architecture: Decentralised ownership of data products using open source tooling.

AI/ML integration: Open ML platforms like MLflow, Kubeflow, and Hugging Face are shaping end-to-end pipelines.

Composable data stacks: Using best-of-breed open tools that interoperate via APIs and standard protocols.

Open source will continue to empower organisations, giving them control over their infrastructure and the ability to innovate at scale. **END** 🐧

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